Algebraic Geometry – Exercise Sheet 1

Week 1: Affine algebraic sets, projective space and localisation

Monday and Thursday afternoon sessions; Lectures 1–5

Difficulty. $\star$ routine; $\star\star$ synthesis; $\star\star\star$ challenge or extension.
Use. The sheet is deliberately longer than the exercise sessions. Cerulean exercises form the suggested route; green exercises are alternatives.
Hints. Read them progressively and stop as soon as one is enough to restart the problem.

Monday afternoon — material available: Lecture 1 only

Exercise 1. Three points in the affine plane $\star$ **CORE**

Let

$$S = \{(0,0),(1,0),(0,1)\} \subset \Bbbk^2.$$

(i) Show that

$$\mathbf{I}(S) = (x,y) \cap (x-1,y) \cap (x,y-1).$$

(ii) Prove directly that

$$\mathbf{I}(S) = \big(xy,\ x(x-1),\ y(y-1)\big).$$

(iii) Show that evaluation at the three points induces an isomorphism

$$\Bbbk[x,y]/\mathbf{I}(S) \simeq \Bbbk^3.$$

Exercise 2. Two axes and their coordinate rings $\star$ **CORE**

In $\Bbbk^2$, set $X = \mathbf{V}(x)$ and $Y = \mathbf{V}(y)$.

(i) Show that

$$X \cup Y = \mathbf{V}(xy), \qquad X \cap Y = \mathbf{V}(x,y).$$

(ii) Compute the coordinate rings of X, Y, $X \cup Y$, and $X \cap Y$.

(iii) Explain geometrically why $\Bbbk[x,y]/(xy)$ has zero divisors.

Exercise 3. The parabola and the cusp $\star\star$ **CORE**

Consider

$$P = \mathbf{V}(y-x^2), \qquad C = \mathbf{V}(y^2-x^3) \subset \Bbbk^2.$$

(i) Show that $t \mapsto (t,t^2)$ defines an isomorphism $\mathbb{A}^1 \to P$.

(ii) Show that $t \mapsto (t^2,t^3)$ defines a morphism $\nu : \mathbb{A}^1 \to C$ which is bijective on the underlying sets of $\Bbbk$-points.

(iii) Write the corresponding homomorphism of coordinate rings and prove that ν is not an isomorphism.

Exercise 4. Polynomial maps of the affine line $\star$ **CHOICE**

Show that morphisms of affine algebraic sets $\mathbb{A}^1_{\Bbbk} \to \mathbb{A}^1_{\Bbbk}$ are exactly polynomial maps $t \mapsto f(t)$. Describe the coordinate-ring homomorphism and check explicitly how composition reverses arrows.

Thursday afternoon — material available: Lectures 1–4

Exercise 5. Radicals and reduced rings $\star$ **CORE**

Compute

$$\sqrt{(x^2,y^3)}, \qquad \sqrt{(x^2y^3)}$$

in $\Bbbk[x,y]$. Decide which of the rings

$$\Bbbk[x]/(x^2), \qquad \Bbbk[x,y]/(xy), \qquad \Bbbk[x,y]/(x^2,xy)$$

are reduced.

Hint 1 (conceptual). Recall that R/I is reduced if and only if I is a radical ideal.

Hint 2 (technical). For $\sqrt{(x^2, y^3)}$, first show that both x and y lie in the radical, then use the prime ideal (x, y) for the reverse inclusion.

Hint 3 (technical). Write x^2y^3 as a product of powers of its distinct irreducible factors. Which multiplicities disappear when one takes the radical? For the quotient rings, try to exhibit a non-zero nilpotent whenever the defining ideal is not radical.

Exercise 6. Points at infinity ⋆ CORE

In the affine chart $x_0 \neq 0$ of $\mathbb{P}^2_{\Bbbk}$, write $x = x_1/x_0$ and $y = x_2/x_0$.

(i) Homogenise the parabola $y = x^2$ and determine its points at infinity.

(ii) Homogenise the hyperbola $xy = 1$ and determine its points at infinity.

(iii) Explain geometrically why the answers are different.

Hint 1 (technical). Homogenise to the largest total degree occurring in the affine equation. Points at infinity are then obtained by adding the equation $x_0 = 0$.

Hint 2 (conceptual). The equation left on the line at infinity is the highest-degree homogeneous part of the original affine equation. Compare its projective zeroes for the parabola and the hyperbola.

Exercise 7. The line through two projective points ⋆ CHOICE

Let $p = [a_0 : \cdots : a_n]$ and $q = [b_0 : \cdots : b_n]$ be distinct points of $\mathbb{P}^n_{\Bbbk}$. Show that the line through them is defined by the 3×3 minors of

$$\begin{pmatrix} a_0 & \cdots & a_n \\ b_0 & \cdots & b_n \\ x_0 & \cdots & x_n \end{pmatrix}.$$

Write the equation explicitly for $n = 2$.

Hint 1 (conceptual). Choose non-zero representatives $a, b, x \in \Bbbk^{n+1}$. The point $[x]$ lies on the line through $[a]$ and $[b]$ precisely when x belongs to the two-dimensional vector space spanned by a and b.

Hint 2 (technical). Translate the preceding condition into a rank condition on the displayed matrix. Recall how rank at most 2 is detected by minors.

Exercise 8. Points in $\mathbb{P}^2$ from minors ⋆⋆ CHOICE

Show that the 2×2 minors of

$$M = \begin{pmatrix} x & 0 \\ y & y \\ 0 & z \end{pmatrix}$$

define the three coordinate points in $\mathbb{P}^2$. For four lines $\ell_1, \ldots, \ell_4$ with no three concurrent, show that the maximal minors of

$$\begin{pmatrix} \ell_1 & 0 & 0 \\ -\ell_2 & \ell_2 & 0 \\ 0 & -\ell_3 & \ell_3 \\ 0 & 0 & -\ell_4 \end{pmatrix}$$

define their six pairwise intersection points.

Suggested strategy. First determine the support by computing the maximal minors. Only afterwards check that the scheme structure at each point is reduced.

Hint 1 (technical). For the first matrix, compute the three minors explicitly. If $xy = xz = yz = 0$ at a projective point, how many coordinates can be non-zero?

Hint 2 (technical). For the second matrix, deleting one row gives a product of three of the four linear forms. Show that all maximal minors vanish exactly when at least two of the ℓ_i vanish.

Hint 3 (technical). Near an intersection point $\ell_i = \ell_j = 0$, the other two linear forms are units because no three lines are concurrent. Use two of the minors to identify the local ideal with (ℓ_i, ℓ_j).

Exercise 9. Equations of the Segre and Veronese images ⋆⋆⋆ OPTIONAL

a) For

$$([s_0 : s_1], [t_0 : t_1]) \longmapsto [s_0t_0 : s_0t_1 : s_1t_0 : s_1t_1],$$

compute the kernel of $z_{ij} \mapsto s_it_j$ and identify the image in $\mathbb{P}^3$.

b) For the quadratic Veronese map $\mathbb{P}^2 \to \mathbb{P}^5$, arrange the target coordinates as a symmetric 3×3 matrix and show that the image is defined by its 2×2 minors.

Suggested strategy. In each case, first find evident determinantal relations. The substantial point is then to prove that these relations generate the whole kernel, not merely that they vanish on the image.

Hint 1 (technical). For the Segre map, two monomials have the same image when their 2×2 exponent tables have the same row and column sums. Try to connect such tables by the elementary move corresponding to $z_{00}z_{11} = z_{01}z_{10}$.

Hint 2 (technical). For the Veronese map, view a monomial in the z_{ij} as a pairing of a multiset of indices. A 2×2 minor changes two pairs $(i,j),(k,\ell)$ into $(i,k),(j,\ell)$.

Hint 3 (technical). To prove that all pairings are connected by these switches, force one chosen pair to agree with the target pairing and argue by induction on the number of remaining pairs.

Exercise 10. Localising the coordinate axes $\star$ **CORE**

Let
$$A = \Bbbk[x,y]/(xy).$$

(i) Compute A_x and A_y.

(ii) Compute $A_{(x)}$ and $A_{(y)}$.

(iii) Explain what these localisations retain and what they discard geometrically.

Hint 1 (conceptual). Keep separate the principal localisation A_x, which describes the open set $D(x)$, and the local ring $A_{(x)}$, which localises at the prime (x).

Hint 2 (technical). In A_x, the element x is invertible, so the relation $xy = 0$ forces $y = 0$. For $A_{(x)}$, every non-zero polynomial in y lies outside (x) and becomes invertible.

Hint 3 (conceptual). The prime (x) is the generic point of the component $V(x)$, which is the y-axis. This explains why its local ring has fraction field $\Bbbk(y)$ rather than $\Bbbk(x)$.

Exercise 11. Localisation with a nilpotent $\star$ **CHOICE**

Let $A = \Bbbk[\varepsilon]/(\varepsilon^2)$. Compute
$$A_\varepsilon, \qquad A_{(\varepsilon)}, \qquad A_{1+\varepsilon}.$$

Explain why the three answers are so different.

Hint 1 (technical). The multiplicative set used to form A_ε contains $\varepsilon^2 = 0$. What happens to a localisation in which 0 becomes invertible?

Hint 2 (technical). Describe the elements outside the prime (ε). Show directly that each $a + b\varepsilon$ with $a \neq 0$ is already a unit.

Hint 3 (technical). Compute $(1+\varepsilon)(1-\varepsilon)$.

Exercise 12. Two finite module structures $\star\star$ **CHOICE**

Let
$$B = \Bbbk[x,y]/(y^2 - x).$$

Show that B is free of rank 2 over $\Bbbk[x]$ and free of rank 1 over $\Bbbk[y]$. Compare geometrically the corresponding morphisms to the affine line.

Hint 1 (technical). Use $x = y^2$. Every polynomial in y has a unique even–odd decomposition
$$f(y) = a(y^2) + yb(y^2).$$

This should produce a basis over $\Bbbk[x]$.

Hint 2 (conceptual). The two algebra structures give two different maps of the same parabola to an affine line: projection to the x-coordinate and parametrisation by the y-coordinate.

Exercise 13. Noether normalisation for the hyperbola $\star\star\star$ **CHOICE**

Let
$$B = \Bbbk[x,y]/(xy-1), \qquad u = x + y \in B.$$

Show that B is finite over $\Bbbk[u]$. Deduce that the corresponding morphism to $\mathbb{A}^1_{\Bbbk}$ has fibres with at most two points.

Suggested strategy. Replace the hyperbola by the Laurent polynomial ring $\Bbbk[x, x^{-1}]$, express u there, and look for a monic equation satisfied by x over $\Bbbk[u]$.

Hint 1 (technical). Since $y = x^{-1}$ and $u = x + x^{-1}$, multiply by x to obtain a quadratic equation for x. Also express x^{-1} in terms of u and x.

Hint 2 (technical). For the fibre over $u = a$, impose the relation $u - a = 0$. The resulting coordinate ring is governed by a polynomial of degree 2 in x.

After Friday's lecture or homework

Exercise 14. A presheaf which is not a sheaf $\star$ OPTIONAL

Let $X = U \amalg V$ be the disjoint union of two non-empty open subsets. Define a presheaf $\mathcal{F}$ by

$$\mathcal{F}(W) = \mathbb{Z} \quad \text{for } W \neq \varnothing, \qquad \mathcal{F}(\varnothing) = 0,$$

with the evident restriction maps. Show that $\mathcal{F}$ is not a sheaf.

Hint 1 (conceptual). Use the cover $X = U \cup V$. Sections over U and V automatically agree on the empty intersection; choose two that cannot be restrictions of one integer over X.

Exercise 15. Continuous and holomorphic functions $\star\star$ CHOICE

Let $U \subset \mathbb{C}$ be connected and contain at least two points. Show that $\mathcal{O}(U)$ has no zero divisors, whereas $\mathcal{C}^0(U)$ has non-zero functions f, g with $fg = 0$.

Hint 1 (conceptual). For holomorphic functions, think of the identity theorem: if one factor is not identically zero, the other vanishes on a non-empty open subset.

Hint 2 (technical). For continuous functions, choose two disjoint small closed discs in U and non-zero continuous functions supported in the two discs.

Exercise 16. $\mathcal{O}(d)$ on the projective line $\star\star\star$ OPTIONAL

Identify $\mathbb{P}^1_{\mathbb{C}}$ with $\mathbb{C} \cup \{\infty\}$. Define $\mathcal{O}(d)$ as the sheaf of meromorphic functions which are holomorphic away from ∞ and have at most a pole of order d there when $d \geq 0$; for $d < 0$, require a zero of order at least $-d$ at ∞. Compute its global sections.

Suggested strategy. Work first on the affine chart $\mathbb{C} = \mathbb{P}^1 \setminus \{\infty\}$, then translate the condition at infinity using the coordinate $w = 1/z$.

Hint 1 (technical). A meromorphic function on $\mathbb{P}^1$ with no finite pole is a polynomial in z. A polynomial of degree m has a pole of order m at infinity.

Hint 2 (technical). When $d < 0$, a section is holomorphic everywhere and must vanish at infinity. Recall the global holomorphic functions on $\mathbb{P}^1$.

Algebraic Geometry – Exercise Sheet 2

Week 2: Prime spectra, the structure sheaf and affine subschemes

Monday and Thursday afternoon sessions; Lectures 6–10

Difficulty. $\star$ routine; $\star\star$ synthesis; $\star\star\star$ challenge or extension.
Use. The sheet is deliberately longer than the exercise sessions. Cerulean exercises form the suggested route; green exercises are alternatives.
Hints. Read them progressively and stop as soon as one is enough to restart the problem.

Monday afternoon — material available: Lecture 6 only

Exercise 1. The affine line as a spectrum $\star$ **CORE**

Assume that $\Bbbk$ is algebraically closed and let $X = \operatorname{Spec}(\Bbbk[t])$.

(i) List the points of X.

(ii) Determine which points are closed and compute the closure of each point.

(iii) For a non-zero polynomial f, describe the closed subset $\mathbb{V}(f)$.

Exercise 2. The spectrum of the integers $\star$ **CORE**

Describe $\operatorname{Spec}(\mathbb{Z})$: its points, its closed points, and the closure of each point. For $n \neq 0$, determine $\mathbb{V}(n)$.

Exercise 3. Nilpotents are invisible topologically $\star$ **CORE**

Let

$$A = \Bbbk[\varepsilon]/(\varepsilon^2), \qquad X = \operatorname{Spec}(A).$$

Determine the points and open subsets of X. Compare the underlying topological space with $\operatorname{Spec}(\Bbbk)$, and explain why the two affine schemes are nevertheless different.

Hint 1 (conceptual). Every prime ideal contains every nilpotent element. Apply this to the class of ε.

Hint 2 (conceptual). The topology only sees prime ideals. To distinguish the two schemes, compare their rings of global sections and ask whether these rings are reduced.

Exercise 4. A quotient map on spectra $\star$ **CHOICE**

Consider the quotient homomorphism

$$\Bbbk[t] \longrightarrow \Bbbk[t]/(t^2).$$

Describe the induced map on spectra and its image. Why does the image depend only on $\sqrt{(t^2)}$?

Hint 1 (conceptual). For a quotient $A \to A/I$, the induced map identifies $\operatorname{Spec}(A/I)$ with the closed subset $V(I)$. Prime ideals contain I exactly when they contain $\sqrt{I}$.

Exercise 5. Closed subsets in the node $\star\star$ **CHOICE**

Let $A = \Bbbk[x, y]/(xy)$ and $X = \operatorname{Spec}(A)$. Denote the classes of x, y by the same letters.

(i) Show that $\mathbb{V}(x) \cup \mathbb{V}(y) = X$ and determine $\mathbb{V}(x) \cap \mathbb{V}(y)$.

(ii) Compute the closures of the points (x) and (y).

(iii) Show that neither of these two points is closed.

Hint 1 (conceptual). A prime ideal containing xy contains x or y. Also recall that the closure of the point $\mathfrak{p}$ in $\operatorname{Spec}(A)$ is $V(\mathfrak{p})$.

Hint 2 (technical). A point $\mathfrak{p}$ is closed precisely when $\mathfrak{p}$ is maximal. Compare (x) and (y) with the maximal ideal (x, y).

Exercise 6. The affine line over a function field $\star\star$ **CHOICE**

Let $K = \mathbb{C}(t)$ and let $X = \operatorname{Spec}(K[x])$.

(i) Describe the points and the closed points of X.

(ii) Which closed points are K-rational? Give a closed point which is not K-rational.

(iii) Put $S = \mathbb{C}[t] \setminus \{0\}$. Use $K[x] = S^{-1}\mathbb{C}[t, x]$ to identify X with the prime ideals of $\mathbb{C}[t, x]$ lying over $(0) \in \operatorname{Spec}(\mathbb{C}[t])$.

Suggested strategy. Treat $K[x]$ exactly as a polynomial ring over an arbitrary field, then use the prime correspondence for the localisation $S^{-1}\mathbb{C}[t, x]$.

Hint 1 (technical). The non-zero prime ideals of the PID $K[x]$ are generated by irreducible polynomials. Such a point is K-rational exactly when its residue field is K.

Hint 2 (technical). To produce a non-rational point, find an irreducible polynomial of degree 2 in $K[x]$; for example, decide whether t is a square in $\mathbb{C}(t)$.

Hint 3 (technical). A prime $\mathfrak{q} \subset \mathbb{C}[t, x]$ survives the localisation precisely when $\mathfrak{q} \cap S = \varnothing$. Rewrite this as a condition on $\mathfrak{q} \cap \mathbb{C}[t]$.

Thursday afternoon — material available: Lectures 6–9

Exercise 7. Principal opens in the node ⋆⋆ **CORE**

Let $A = \Bbbk[x, y]/(xy)$ and $X = \operatorname{Spec}(A)$.

(i) Describe $D(x)$ and $D(y)$ as affine schemes.

(ii) Show that $D(x) \cap D(y) = \varnothing$ and identify the point missing from $D(x) \cup D(y)$.

(iii) Compute the residue fields at (x), (y) and (x, y).

Hint 1 (technical). Compute A_x and A_y first. The equation $xy = 0$ simplifies drastically after either x or y is inverted.

Hint 2 (technical). Use $D(x) \cap D(y) = D(xy)$ and remember that $xy = 0$ in A.

Hint 3 (technical). For a prime $\mathfrak{p}$, use

$$\kappa(\mathfrak{p}) = \operatorname{Frac}(A/\mathfrak{p}).$$

Apply this separately to (x), (y) and (x, y).

Exercise 8. Sections and a stalk on the affine line ⋆ **CORE**

Let $X = \operatorname{Spec}(\Bbbk[t])$ and $U = D(t(t-1))$.

(i) Compute $\mathcal{O}_X(U)$.

(ii) Compute the stalk $\mathcal{O}_{X,(t-a)}$ and its residue field.

(iii) For which a does $(t-a)$ belong to U?

Hint 1 (technical). For an affine scheme, $\Gamma(D(f), \mathcal{O}) = A_f$. Here $f = t(t-1)$.

Hint 2 (technical). The point $(t-a)$ belongs to $D(f)$ if and only if $f \notin (t-a)$, equivalently if and only if $f(a) \neq 0$.

Exercise 9. A subscheme of the affine line with two support points ⋆⋆ **CORE**

Let

$$Z = \operatorname{Spec}\bigl(\Bbbk[t]/(t^3(t-1)^2)\bigr).$$

(i) Determine the underlying topological space of Z.

(ii) Decompose its coordinate ring as a product of two rings.

(iii) Compare the two local pieces with the reduced points at 0 and 1.

Suggested strategy. First take the radical to find the support. Then separate the two coprime primary factors by the Chinese remainder theorem.

Hint 1 (technical). The ideals (t^3) and $((t-1)^2)$ are comaximal. Apply the Chinese remainder theorem before interpreting the two factors geometrically.

Hint 2 (conceptual). Each factor has the topology of one reduced point, but its maximal ideal contains nilpotents. Compare their vector-space lengths.

Exercise 10. Scheme-theoretic unions and intersections ⋆ **CORE**

In $\mathbb{A}^2_{\Bbbk}$, let
$$X = \mathbb{V}(x), \qquad Y = \mathbb{V}(y), \qquad Y' = \mathbb{V}(y^2).$$
Compute the ideals defining $X \cup Y$, $X \cap Y$, $X \cup Y'$ and $X \cap Y'$. Compare their underlying closed subsets.

Hint 1 (conceptual). For closed subschemes of an affine scheme, union corresponds to intersection of ideals and intersection corresponds to sum of ideals.

Hint 2 (technical). Compute $(x) \cap (y)$ and $(x) \cap (y^2)$ using divisibility in the UFD $\Bbbk[x, y]$. Take radicals only after finding the scheme-theoretic ideals.

Exercise 11. Products of rings and disjoint unions ⋆⋆ **CHOICE**

Let B and C be non-zero rings.

(i) Show that every prime ideal of $B \times C$ is of the form $\mathfrak{p} \times C$ or $B \times \mathfrak{q}$.

(ii) Deduce that
$$\mathrm{Spec}(B \times C) \simeq \mathrm{Spec}(B) \amalg \mathrm{Spec}(C).$$

(iii) Relate this to the idempotents $(1, 0)$ and $(0, 1)$.

Hint 1 (conceptual). Use the complementary idempotents $e = (1, 0)$ and $1 - e = (0, 1)$. Since $e(1 - e) = 0$, a prime ideal must contain one of them.

Hint 2 (technical). If $(1, 0)$ lies in a prime P, then $B \times 0 \subset P$. Identify the remaining part of P with a prime ideal of C, and argue symmetrically in the other case.

Hint 3 (conceptual). The opens $D(1, 0)$ and $D(0, 1)$ are disjoint, open and closed, and cover the spectrum.

After Friday's arithmetic lecture or homework

Exercise 12. Prime ideals in $\mathbb{Z}[\sqrt{3}]$ ⋆⋆ **CHOICE**

Let
$$A = \mathbb{Z}[s]/(s^2 - 3), \qquad f : \mathrm{Spec}(A) \longrightarrow \mathrm{Spec}(\mathbb{Z}).$$
For a prime p, describe the points of $\mathrm{Spec}(A)$ mapping to (p) by factoring $s^2 - 3$ over $\mathbf{F}_p$. Treat separately $p = 2$, $p = 3$, and $p \neq 2, 3$.

Suggested strategy. Compute the fibre ring over (p) and read its points from the irreducible factorisation of $s^2 - 3$ in $\mathbf{F}_p[s]$.

Hint 1 (technical). Treat $p = 2$ and $p = 3$ directly: the polynomial has a repeated root in each case.

Hint 2 (technical). For $p \neq 2, 3$, distinguish whether 3 is a quadratic residue modulo p. In the irreducible case, determine the residue field of the unique point.

Exercise 13. An arithmetic conic ⋆⋆⋆ **OPTIONAL**

For
$$X = \mathrm{Spec}\big(\mathbb{Z}[x, y]/(x^2 + y^2 - 5)\big) \longrightarrow \mathrm{Spec}(\mathbb{Z}),$$
write down the equation obtained after reduction modulo a prime p. Determine for which p that equation is a square or has a singular point.

Suggested strategy. Analyse the fibre over each prime by reducing the equation modulo p. For odd p, use the partial derivatives to locate singular points; handle characteristic 2 separately.

Hint 1 (technical). For odd p, a singular point must satisfy $2x = 2y = 0$ as well as the equation. Substitute the resulting candidate into $x^2 + y^2 - 5$.

Hint 2 (technical). In characteristic 2, use $(x + y + 1)^2 = x^2 + y^2 + 1$. At $p = 5$, decide whether -1 is a square and factor the quadratic form.

Algebraic Geometry – Exercise Sheet 3

Week 3: Schemes, morphisms, gluing and first global properties

Monday and Thursday afternoon sessions; Lectures 11–15

Difficulty. $\star$ routine; $\star\star$ synthesis; $\star\star\star$ challenge or extension.
Use. The sheet is deliberately longer than the exercise sessions. Cerulean exercises form the suggested route; green exercises are alternatives.
Hints. Read them progressively and stop as soon as one is enough to restart the problem.

Monday afternoon — material available: Lectures 1–11

Exercise 1. Global functions as morphisms to the affine line $\star$ **CORE**

Let X be a scheme.

(i) Explain why a global section $s \in \Gamma(X, \mathcal{O}_X)$ determines a morphism

$$f_s : X \longrightarrow \mathbb{A}^1_{\mathbb{Z}}.$$

(ii) Show that

$$\mathrm{Hom}(X, \mathbb{A}^1_{\mathbb{Z}}) \simeq \Gamma(X, \mathcal{O}_X).$$

(iii) If $X = \mathrm{Spec}(A)$, describe the morphism associated with $a \in A$.

Hint 1 (conceptual). Use the affine-target adjunction for $\mathbb{A}^1_{\mathbb{Z}} = \mathrm{Spec}(\mathbb{Z}[t])$.

Hint 2 (technical). A unital ring homomorphism $\mathbb{Z}[t] \to \Gamma(X, \mathcal{O}_X)$ is uniquely determined by the image of t.

Exercise 2. The reduced point and the double point $\star$ **CORE**

Inside $\mathbb{A}^1_{\Bbbk} = \mathrm{Spec}(\Bbbk[t])$, consider

$$P = \mathrm{Spec}(\Bbbk[t]/(t)), \qquad D = \mathrm{Spec}(\Bbbk[t]/(t^2)).$$

(i) Show that the two closed immersions have the same image as subsets of $\mathbb{A}^1_{\Bbbk}$.

(ii) Show that P and D are not isomorphic as schemes.

(iii) Compute the maps on local rings at their unique points.

Hint 1 (conceptual). The two images as subsets are $V(t)$ and $V(t^2)$. Compare their radicals.

Hint 2 (technical). Each source has only one point, so its local ring there is its whole coordinate ring. Reducedness already distinguishes the two schemes.

Exercise 3. A morphism between affine schemes $\star$ **CORE**

Let

$$\varphi : \Bbbk[u, v] \longrightarrow \Bbbk[t], \qquad u \longmapsto t, \quad v \longmapsto t^2.$$

(i) Describe the induced morphism $f : \mathbb{A}^1_{\Bbbk} \to \mathbb{A}^2_{\Bbbk}$.

(ii) Determine $\ker(\varphi)$ and factor f through a closed subscheme of $\mathbb{A}^2_{\Bbbk}$.

(iii) Show that the resulting morphism onto that closed subscheme is an isomorphism.

Hint 1 (technical). The relation $v - u^2$ is in the kernel. Divide an arbitrary $F(u, v)$ by the monic polynomial $v - u^2$, viewing it as a polynomial in v.

Hint 2 (technical). After passing to $\Bbbk[u, v]/(v - u^2)$, write down an inverse to the induced map by specifying the image of t.

Exercise 4. An arithmetic spectrum $\star\star$ **CHOICE**

Let $A = \mathbb{Z}[s]/(s^2 - 3)$. Describe the inverse image in $\mathrm{Spec}(A)$ of the points (2), (3), and (p) for $p \neq 2, 3$, according to the factorisation of $s^2 - 3$ modulo p.

Hint 1 (technical). The inverse image of (p) is the spectrum of $\mathbf{F}_p[s]/(s^2 - 3)$. Factor the polynomial in the three cases $p = 2$, $p = 3$, and $p \neq 2, 3$.

Thursday afternoon — material available: Lectures 1–14

Exercise 5. Dual numbers and tangent vectors ⋆⋆ CORE

Let $D = \operatorname{Spec}(\Bbbk[\varepsilon]/(\varepsilon^2))$ and let $p = (a, b) \in \mathbb{A}^2_{\Bbbk}(\Bbbk)$.

(i) Show that morphisms $D \to \mathbb{A}^2_{\Bbbk}$ whose closed point maps to p are uniquely of the form

$$x \longmapsto a + u\varepsilon, \qquad y \longmapsto b + v\varepsilon.$$

(ii) Let $C = \mathbb{V}(f) \subset \mathbb{A}^2_{\Bbbk}$. Show that such a morphism factors through C precisely when

$$f(a, b) = 0, \qquad f_x(a, b)u + f_y(a, b)v = 0.$$

(iii) Apply this to the parabola $y = x^2$.

Suggested strategy. Translate the morphism into a homomorphism of coordinate rings, then expand f after substituting $a + u\varepsilon$ and $b + v\varepsilon$ and discard every term containing ε^2.

Hint 1 (technical). Every element of $\Bbbk[\varepsilon]/(\varepsilon^2)$ is uniquely $c + d\varepsilon$. Mapping the closed point to (a, b) fixes the constant terms of the images of x and y.

Hint 2 (technical). Prove the first-order formula

$$f(a + u\varepsilon, b + v\varepsilon) = f(a, b) + \big(f_x(a, b)u + f_y(a, b)v\big)\varepsilon.$$

Exercise 6. The line with two origins ⋆⋆ CORE

Glue two copies

$$X_1 = \operatorname{Spec}(\Bbbk[t_1]), \qquad X_2 = \operatorname{Spec}(\Bbbk[t_2])$$

along $D(t_1) \simeq D(t_2)$ by $t_1 \mapsto t_2$. Let X be the resulting scheme.

(i) Describe the underlying topological space.

(ii) Show that $\Gamma(X, \mathcal{O}_X) \simeq \Bbbk[t]$.

(iii) Show that the canonical map $X \to \operatorname{Spec}\Gamma(X, \mathcal{O}_X)$ identifies the two origins.

(iv) Deduce that X is not affine.

Suggested strategy. Describe the gluing topologically first. Then compute global functions as compatible pairs on the two affine charts. Finally compare the scheme with the spectrum of that global ring.

Hint 1 (technical). A global function is a pair $(f_1, f_2) \in \Bbbk[t_1] \times \Bbbk[t_2]$ whose images agree in $\Bbbk[t, t^{-1}]$. Use the injectivity of $\Bbbk[t] \to \Bbbk[t, t^{-1}]$.

Hint 2 (conceptual). The canonical map to $\operatorname{Spec}\Gamma(X, \mathcal{O}_X)$ restricts to the ordinary coordinate map on both charts. Track the two origins under this map.

Exercise 7. Global functions on the projective line ⋆⋆ CHOICE

Construct $\mathbb{P}^1_{\Bbbk}$ by gluing $U_0 = \operatorname{Spec}(\Bbbk[t])$ and $U_1 = \operatorname{Spec}(\Bbbk[s])$ along $s = t^{-1}$. Prove that

$$\Gamma(\mathbb{P}^1_{\Bbbk}, \mathcal{O}_{\mathbb{P}^1}) = \Bbbk.$$

Hint 1 (technical). A global function is a pair $f(t) \in \Bbbk[t]$, $g(s) \in \Bbbk[s]$ satisfying $f(t) = g(t^{-1})$ in $\Bbbk[t, t^{-1}]$. Compare the possible positive and negative powers of t.

Exercise 8. The projective conic from its standard opens ⋆⋆ CORE

Let

$$S = \Bbbk[x_0, x_1, x_2]/(x_0x_2 - x_1^2), \qquad C = \operatorname{Proj}(S).$$

(i) Compute $(S_{x_0})_0$ and $(S_{x_2})_0$.

(ii) Identify $D_+(x_0)$ and $D_+(x_2)$ with affine lines.

(iii) Describe their overlap and recover the gluing rule for $\mathbb{P}^1$.

Suggested strategy. Compute the degree-zero parts of the two homogeneous localisations, then express the transition function on the overlap.

Hint 1 (technical). On $D_+(x_0)$ use $t = x_1/x_0$ and the relation $x_2/x_0 = t^2$. Do the symmetric computation on $D_+(x_2)$.

Hint 2 (technical). On the overlap, both x_0 and x_2 are non-zero. Compare the two affine parameters using $x_0x_2 = x_1^2$.

Exercise 9. Connectedness and idempotents ⋆⋆ CORE

(i) Prove that $\mathrm{Spec}(A)$ is disconnected if and only if A has a non-trivial idempotent.

(ii) Apply this to $A = \Bbbk[t]/(t^2 - t)$.

(iii) Show that $\mathrm{Spec}(\Bbbk[x, y]/(xy))$ is connected although reducible.

Hint 1 (conceptual). A decomposition of $\mathrm{Spec}(A)$ into two disjoint open-and-closed subsets should correspond to a function taking the values 0 and 1 on the two pieces.

Hint 2 (technical). Starting from an idempotent e, use the two principal opens $D(e)$ and $D(1 - e)$. Conversely, translate a clopen decomposition into a product decomposition of A.

Hint 3 (technical). For $\Bbbk[x, y]/(xy)$, an idempotent restricts to idempotents on both axes. Compare their values at the common origin.

Exercise 10. Reduction and its universal property ⋆⋆ CORE

Let

$$A = \Bbbk[x, y]/(x^2, xy), \qquad X = \mathrm{Spec}(A).$$

(i) Compute the nilradical and X_{red}.

(ii) Let R be a reduced $\Bbbk$-algebra. Show that every homomorphism $A \to R$ factors uniquely through A_{red}.

(iii) Translate this into a statement about morphisms of schemes.

Hint 1 (conceptual). Find the nilradical first. If R is reduced, the image in R of every nilpotent element of A must vanish.

Hint 2 (technical). Use this to show that the kernel of any homomorphism $A \to R$ contains the nilradical. The desired factorisation is then the universal property of a quotient ring.

Hint 3 (conceptual). Remember to reverse arrows when translating the factorisation into a statement about schemes.

Exercise 11. A weighted projective chart ⋆⋆⋆ OPTIONAL

Let $R = \Bbbk[x, y, z]$ with $\deg x = \deg y = 1$ and $\deg z = 2$, and set $X = \mathrm{Proj}(R)$. Show that $D_+(x)$ and $D_+(y)$ are affine planes and that

$$D_+(z) \simeq \mathrm{Spec}\big(\Bbbk[a, b, c]/(ac - b^2)\big).$$

Identify the singular point of this chart.

Suggested strategy. For each standard open $D_+(f)$, compute the degree-zero subring $(R_f)_0$. The chart at z is the only one where the grading produces a relation.

Hint 1 (technical). On $D_+(z)$ consider the degree-zero elements

$$a = x^2/z, \qquad b = xy/z, \qquad c = y^2/z.$$

They satisfy one evident relation.

Hint 2 (technical). To locate the singular point of $ac - b^2 = 0$, use the partial derivatives with respect to a, b, c.

After Friday's lecture or preparation for Week 4

Exercise 12. Components and generic points ⋆⋆ CHOICE

Let

$$I = (xy, xz) \subset \Bbbk[x, y, z], \qquad X = \operatorname{Spec}(\Bbbk[x, y, z]/I).$$

Show that $I = (x) \cap (y, z)$. Determine the irreducible components, their generic points and dimensions, and decide whether X is connected or integral.

Hint 1 (conceptual). For an affine scheme, irreducible components correspond to minimal prime ideals, and their generic points are those minimal primes viewed as points of the spectrum.

Hint 2 (technical). To prove $(x) \cap (y, z) = (xy, xz)$, write an element of the intersection as xg and reduce modulo (y, z).

Hint 3 (conceptual). Use the definition of irreducibility and inspect whether the two components meet. Reducedness comes from the expression of I as an intersection of prime ideals.

Exercise 13. The function field of the cusp ⋆⋆ **CHOICE**

Let

$$C = \operatorname{Spec}\big(\Bbbk[x, y]/(y^2 - x^3)\big).$$

Show that C is integral and that $K(C) \simeq \Bbbk(t)$.

Hint 1 (technical). Use the map $x \mapsto t^2$, $y \mapsto t^3$. Every class in the coordinate ring has a representative $A(x) + yB(x)$; parity of exponents proves injectivity.

Hint 2 (technical). The fraction field is contained in $\Bbbk(t)$. Recover t as a quotient of the images of x and y.

Algebraic Geometry – Exercise Sheet 4

Week 4: Generic points, dimension and scheme-theoretic fibres

One Monday exercise session, followed by a revision bank; Lectures 15–18

Difficulty. $\star$ routine; $\star\star$ synthesis; $\star\star\star$ challenge or extension.
Use. The sheet is deliberately longer than the exercise sessions. Cerulean exercises form the suggested route; green exercises are alternatives.
Hints. Read them progressively and stop as soon as one is enough to restart the problem.

Monday afternoon — material available: Lectures 1–16

Exercise 1. Components and generic points $\star\star$ **CORE**

Let

$$I = (xy, xz) \subset \Bbbk[x, y, z], \qquad X = \operatorname{Spec}(\Bbbk[x, y, z]/I).$$

(i) Show that $I = (x) \cap (y, z)$.

(ii) Determine the irreducible components and their generic points.

(iii) Is X reduced, connected, irreducible, or integral?

Hint 1 (conceptual). For an affine scheme, generic points of irreducible components correspond to minimal prime ideals.

Hint 2 (technical). To prove $(x) \cap (y, z) = (xy, xz)$, start with $f = xg$ in the intersection and reduce the equality modulo (y, z).

Hint 3 (conceptual). An intersection of prime ideals is radical. For connectedness, decide whether the two components meet; for irreducibility, return to the definition.

Exercise 2. The function field of the cusp $\star\star$ **CORE**

Let

$$C = \operatorname{Spec}\big(\Bbbk[x, y]/(y^2 - x^3)\big).$$

Show that C is integral, compute its function field, and deduce its dimension.

Suggested strategy. Embed the coordinate ring in $\Bbbk[t]$, prove that the embedding is injective, and then compare fraction fields.

Hint 1 (technical). Reduce every class modulo $y^2 - x^3$ to $A(x) + yB(x)$. After substituting $x = t^2$, $y = t^3$, separate even and odd powers of t.

Hint 2 (technical). The fraction field contains y/x. What does this element become in $\Bbbk(t)$?

Exercise 3. Chains and dimensions $\star$ **CORE**

(i) Exhibit a maximal chain of irreducible closed subsets in $\operatorname{Spec}(\mathbb{Z})$.

(ii) Give a chain of length n in $\mathbb{A}^n_{\Bbbk}$.

(iii) Determine the dimension of

$$\operatorname{Spec}\big(\Bbbk[x, y, z]/(xy, xz)\big)$$

from its components.

Hint 1 (conceptual). Chains of irreducible closed subsets correspond, with inclusions reversed, to chains of prime ideals.

Hint 2 (technical). In affine space, successively impose $x_1 = 0$, then $x_2 = 0$, and so on. For the last scheme, take the maximum of the dimensions of its irreducible components.

Exercise 4. Noether normalisation and dimension $\star\star$ **CORE**

Let

$$B = \Bbbk[x, y]/(y^2 - x^3).$$

(i) Show that B is finite over $\Bbbk[x]$.

(ii) Use this to recover $\dim(B) = 1$.

(iii) Compare with the computation from the function field.

Hint 1 (technical). The element y satisfies a monic polynomial over $\Bbbk[x]$. Use division by that monic polynomial to generate B as a $\Bbbk[x]$-module.

Hint 2 (conceptual). Recall that a finite integral extension preserves Krull dimension. Compare the result with $\operatorname{trdeg}_{\Bbbk} K(C)$.

After Lectures 17–18 — revision bank on fibres

Exercise 5. The generic affine line ⋆ **CORE**

Let

$$\pi : \mathbb{A}^2_{\mathbb{C}} = \operatorname{Spec}(\mathbb{C}[t,x]) \longrightarrow \mathbb{A}^1_{\mathbb{C}} = \operatorname{Spec}(\mathbb{C}[t])$$

be the projection. Compute its fibre over a closed point $(t-a)$ and its fibre over the generic point (0). Explain why the latter is $\operatorname{Spec}(\mathbb{C}(t)[x])$ and verify the dimension formula for the generic fibre.

Hint 1 (conceptual). The fibre over a point $s \in \operatorname{Spec}(A)$ is obtained by tensoring with its residue field $\kappa(s)$.

Hint 2 (technical). For the closed point $(t-a)$, tensor with $\mathbb{C}[t]/(t-a)$. For the generic point (0), tensor with $\kappa((0)) = \mathbb{C}(t)$.

Exercise 6. A fibre product larger than the set-theoretic one ⋆⋆ **CHOICE**

Let $X = \operatorname{Spec}(\mathbb{C})$ and $S = \operatorname{Spec}(\mathbb{R})$. Compute $X \times_S X$ and compare its points with the set-theoretic fibre product of the underlying one-point spaces.

Hint 1 (technical). For affine schemes, compute the tensor product $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}$. Present the first copy as $\mathbb{R}[T]/(T^2+1)$ and then extend scalars to $\mathbb{C}$.

Hint 2 (technical). Factor T^2+1 over $\mathbb{C}$ and apply the Chinese remainder theorem.

Exercise 7. The power map and its fibres ⋆⋆ **CORE**

Assume that $\Bbbk$ is algebraically closed of characteristic 0. Let

$$f : \mathbb{A}^1_{\Bbbk} \longrightarrow \mathbb{A}^1_{\Bbbk}, \qquad t \longmapsto t^n.$$

Compute every scheme-theoretic fibre, determine when it is reduced, compute the generic fibre, and show that f is finite.

Suggested strategy. Write the coordinate-ring map using a target coordinate s. Compute closed fibres by imposing $s = a$, then treat the generic point by replacing $\Bbbk[s]$ with $\Bbbk(s)$.

Hint 1 (technical). The fibre over a has coordinate ring $\Bbbk[t]/(t^n - a)$. Separate $a = 0$ from $a \neq 0$ and use the derivative of $t^n - a$ to test repeated roots.

Hint 2 (technical). For the generic fibre, study $t^n - s$ over $\Bbbk(s)$. Eisenstein at the prime s in $\Bbbk[s]$ proves irreducibility.

Hint 3 (technical). To prove finiteness, divide exponents modulo n: express each polynomial uniquely as

$$\sum_{r=0}^{n-1} t^r a_r(t^n).$$

Exercise 8. A family whose special fibre becomes reducible ⋆⋆ **CORE**

Let

$$X = \operatorname{Spec}\big(\Bbbk[x,y,t]/(xy-t)\big) \longrightarrow \mathbb{A}^1_t.$$

Compute the fibres, show that all have dimension 1, and determine which are integral and which are reducible.

Hint 1 (technical). The fibre over a is obtained by setting $t = a$, hence has equation $xy = a$.

Hint 2 (technical). If $a \neq 0$, one coordinate is invertible and the ring is a Laurent polynomial ring. If $a = 0$, factor the equation and identify the two components.

Hint 3 (conceptual). Irreducibility can change in a family even when the fibre dimension remains constant.

Exercise 9. A jumping fibre ⋆⋆ CORE

Consider
$$f : \mathbb{A}^2_{\Bbbk} \longrightarrow \mathbb{A}^2_{\Bbbk}, \qquad (x, y) \longmapsto (x, xy).$$
Compute the fibre over (a, b) and show that the generic fibre has dimension 0, while the fibre over $(0, 0)$ has dimension 1.

Suggested strategy. Write the fibre ring over (a, b), eliminate x using $x - a$, and then distinguish the cases $a \neq 0$, $a = 0, b \neq 0$, and $(a, b) = (0, 0)$.

Hint 1 (technical). The fibre ring is
$$\Bbbk[x, y]/(x - a, xy - b).$$
After substituting $x = a$, the remaining equation is $ay = b$.

Hint 2 (technical). For the generic fibre, work over $\Bbbk(u, v)$: the relation $x = u$ makes x invertible and determines y.

Exercise 10. The generic fibre formula in examples ⋆⋆ CHOICE

Verify
$$\dim X_\eta = \dim X - \dim Y$$
for the map in the preceding exercise and for the projection $\mathbb{A}^{m+n}_{\Bbbk} \to \mathbb{A}^m_{\Bbbk}$.

Hint 1 (conceptual). Compute the generic fibre using function fields. Its dimension is the transcendence degree of $K(X)$ over $K(Y)$.

Hint 2 (technical). For $(x, y) \mapsto (x, xy)$, recover x and y rationally from the target coordinates $u = x$ and $v = xy$.

Optional outreach exercises: Hilbert schemes of points

Exercise 11. $\mathrm{Hilb}^n(\mathbb{A}^1)$ ⋆⋆ CHOICE

Assume that $\Bbbk$ is algebraically closed. Show that every ideal $I \subset \Bbbk[x]$ with $\dim_{\Bbbk} \Bbbk[x]/I = n$ is generated by a unique monic polynomial of degree n. Deduce
$$\mathrm{Hilb}^n(\mathbb{A}^1_{\Bbbk}) \simeq \mathbb{A}^n_{\Bbbk}$$
and explain how factorisation records support and multiplicities.

Suggested strategy. First classify finite-colength ideals of the PID $\Bbbk[x]$. Then identify a monic degree-n polynomial with its n coefficients. Keep the functorial argument separate from the classification of $\Bbbk$-points.

Hint 1 (technical). If $I = (f)$ with f monic of degree d, Euclidean division gives the basis $1, x, \ldots, x^{d-1}$ of $\Bbbk[x]/(f)$.

Hint 2 (conceptual). For the scheme-level statement over a ring R, multiplication by x on the rank-n quotient satisfies its characteristic polynomial by Cayley–Hamilton.

Hint 3 (technical). Over an algebraically closed field, factor the monic polynomial. Distinct roots give the support and their exponents give the local lengths.

Exercise 12. Commuting matrices and a cyclic vector ⋆⋆⋆ OPTIONAL

Let V be n-dimensional and let (X, Y, v) satisfy $[X, Y] = 0$ and $\Bbbk[X, Y]v = V$. Show that
$$I = \{f \in \Bbbk[x, y] \mid f(X, Y)v = 0\}$$
is an ideal and $\Bbbk[x, y]/I \simeq V$. Determine the triple for $I = (y, x^2)$.

Suggested strategy. Use the commuting pair to make V into a $\Bbbk[x, y]$-module, with x acting by X and y by Y. The cyclic vector then defines a surjective module map from $\Bbbk[x, y]$ to V.

Hint 1 (technical). Commutativity is exactly what makes $f \mapsto f(X, Y)$ an algebra homomorphism. Use it to show that the annihilator of v is an ideal.

Hint 2 (technical). For $I = (y, x^2)$, use the basis $(1, x)$ of the quotient and compute multiplication by x and by y in this basis.